

RabbitVideo: Fast Diffusion Video Generation on Resource-Constrained Devices

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Abstract

Diffusion models excel at video generation but require substantial computational resources, limiting deployment on consumer-grade GPUs. Unlike large language models, video diffusion models (VDMs) cannot reuse key-value caches across denoising steps, making traditional offloading strategies ineffective and creating a locality mismatch with heterogeneous architectures. We present RabbitVideo, an efficient VDM inference system for local deployments using single consumer GPUs. RabbitVideo exploits temporal sparsity to enable data reuse across heterogeneous memory hierarchies, reducing computation requirements. It leverages spatial sparsity in attention and MLP layers to minimize PCIe data transfers between CPU and GPU. Additionally, RabbitVideo dynamically evaluates component importance to skip unnecessary sublayer executions, reducing both computation and data transfer costs. Evaluation demonstrates that RabbitVideo achieves up to $x \times x \times$ speedup in video generation while maintaining quality within acceptable bounds.

Keywords: Do, Not, Use, This, Code, Put, the, Correct, Terms, for, Your, Paper

1 Introduction

Diffusion models have achieved remarkable success in sophisticated generation tasks, particularly in image and video synthesis. While these models are typically deployed on high-end server-grade GPUs in data centers, there is growing demand to run video diffusion models (VDMs) on resource-constrained platforms with consumer-grade GPUs. This shift is driven by critical needs for data privacy, model customization, and cost reduction in applications ranging from personalized content creation to edge-based video processing.

However, deploying Diffusion Transformer (DiT) models on consumer GPUs faces fundamental challenges that stem from the unique characteristics of video generation. First, computational intensity scales poorly: generating even moderate-quality videos requires processing millions of tokens representing frames across time. The self-attention mechanism’s quadratic complexity means that computational requirements grow rapidly with video resolution and length. Second, memory footprint explodes: the attention sublayer produces substantial intermediate activations, particularly key-value (KV) pairs, whose memory requirements scale linearly with token count. Together, these factors create a perfect storm for resource-constrained deployments, where both compute and memory are severely limited.

Existing techniques [] for reducing resource requirements in large language models (LLMs) appear promising at first glance but fail to address VDM challenges. LLM inference systems achieve efficiency by offloading model weights and KV caches to CPU DRAM, then loading them back during computation. Critically, these systems exploit temporal locality: in LLM autoregressive decoding, input context remains unchanged, allowing KV caches to be computed once and reused across generation steps. This transforms the problem from compute-bound to memory-bound, making offloading effective.

VDMs, however, exhibit fundamentally different behavior. The diffusion process progressively denoises the input across multiple timesteps, with each denoising step producing a different noisy input. This changing input invalidates the KV cache at every step—there is no temporal reuse opportunity. Consequently, offloading model weights merely addresses memory capacity constraints without reducing computational requirements, which remain the primary bottleneck on resource-constrained GPUs. The mismatch between heterogeneous system architectures (designed for locality exploitation) and VDM inference patterns (which lack such locality) renders traditional offloading strategies ineffective.

In this paper, we argue that despite the apparent lack of locality in the diffusion process, VDMs exhibit rich sparsity patterns that can be exploited to match heterogeneous hardware characteristics. We observe two critical forms of sparsity: (1) temporal sparsity across diffusion timesteps, where certain computations and data exhibit predictable reuse patterns despite changing inputs, and (2) spatial sparsity within attention and MLP layers, where many computations contribute minimally to generation quality.

Building on these insights, we introduce RabbitVideo, an efficient VDM inference system optimized for local deployment on single consumer-grade GPUs. Our system makes three key technical contributions:

Contribution 1: Temporal locality exploitation. We identify and leverage temporal sparsity patterns across diffusion steps to enable data reuse across the memory hierarchy, substantially reducing redundant computation even as inputs change.

Contribution 2: Spatial sparsity-aware offloading. We exploit spatial sparsity in attention and MLP layers to dynamically minimize data movement across the slow PCIe interconnect, reducing both memory pressure and transfer overhead.

Contribution 3: Adaptive layer execution. We dynamically evaluate the importance of different DiT components

and selectively skip sublayer execution, reducing both computational cost and data transfer overhead without sacrificing generation quality.

Our evaluation demonstrates that RabbitVideo achieves $x\times$ speedup over baseline implementations while maintaining xx on consumer GPUs with xx , enabling practical local deployment of high-quality VDMs.

The evaluation shows that RabbitVideo can achieve the video generation speed up up to $x\times$ in video generation while keeping a video quality drop within xx .

References